# PropRent

Baserow Database Implementation Specification

Field definitions, recommended Baserow field types, relationships and implementation notes

Purpose: Handoff guide for creating the 20 logical ERD tables in Baserow.

Important: This is a Baserow implementation mapping, not a SQL schema. PostgreSQL was illustrative in the DBML and is not a confirmed database-vendor decision.

## Implementation principles

- Use the table and field names shown here. Use Link to table for ERD foreign-key relationships rather than storing a plain numeric ID for the same relationship.
- Use Single select for statuses/categories and agree allowed options before entering production data.
- Use Number fields for money and permit decimal values. Validate non-negative amounts where required.
- Baserow primary-field behavior may differ from a traditional database PK. Keep the default primary field if needed, and add the named *_id Autonumber field as the logical record identifier.
- Optional links should remain optional. Uniqueness, CHECK constraints, append-only history and date-overlap rules may need workflow/application validation.

# 1. Tables and Fields

## 1. Table: Role

Define user roles.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| role_id | Autonumber | Record identifier. |
| role_name | Single line text | Required; agree role names with the team. |
| description | Long text | Optional description. |

## 2. Table: User

Store user profiles.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| user_id | Autonumber | Record identifier. |
| role_id | Link to table → Role | Required; current model assumes one primary role per user. |
| full_name | Single line text | Required. |
| email | Email | Required; validate uniqueness. |
| phone | Phone number or Single line text | Optional. |
| status | Single select | Account status; agree allowed options. |
| activation_status | Single select | Activation state; agree allowed options. |
| created_at | Created on or Date and time | Use Created on for automatic creation time. |
| updated_at | Last modified or Date and time | Use Last modified for automatic update time. |

## 3. Table: Property

Properties managed by the platform.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| property_id | Autonumber | Record identifier. |
| owner_id | Link to table → User | Property owner. |
| manager_id | Link to table → User | Property manager. |
| name | Single line text | Required. |
| address | Long text | Required. |
| property_type | Single select | Property type. |

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| status | Single select | Property status. |
| created_at | Created on or Date and time | Creation time. |

## 4. Table: Unit

Units belonging to a property.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| unit_id | Autonumber | Record identifier. |
| property_id | Link to table → Property | Required. |
| unit_number | Single line text | Unit identifier within property. |
| bedrooms | Number | Number of bedrooms. |
| bathrooms | Number | Number of bathrooms. |
| rent_amount | Number | Allow decimal values; validate non-negative amount. |
| deposit_amount | Number | Allow decimal values; validate non-negative amount. |
| availability_date | Date | Optional. |
| status | Single select | Unit status. |
| description | Long text | Optional. |

## 5. Table: ListingMedia

Store references to listing images/media.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| media_id | Autonumber | Record identifier. |
| unit_id | Link to table → Unit | Required. |
| storage_key | Single line text | Reference/key to stored file, not necessarily the file itself. |
| media_type | Single select | Media type. |
| uploaded_by | Link to table → User | Uploader. |
| created_at | Date and time | Creation/upload time. |

## 6. Table: ApplicantEnquiry

Track prospective tenant enquiries.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| enquiry_id | Autonumber | Record identifier. |
| unit_id | Link to table → Unit | Required. |
| applicant_name | Single line text | Required. |
| email | Email | Required. |
| phone | Phone number or Single line text | Optional. |
| message | Long text | Optional message. |
| status | Single select | Enquiry status. |
| created_at | Date and time | Creation time. |

## 7. Table: Lease

Rental lease records.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| lease_id | Autonumber | Record identifier. |
| unit_id | Link to table → Unit | Required. |
| tenant_id | Link to table → User | Required; current model assumes one tenant per lease. |
| start_date | Date | Required. |
| end_date | Date | Required; must not precede start date. |
| rent_amount | Number | Allow decimal values. |
| deposit_amount | Number | Allow decimal values. |
| billing_day | Number | Billing day; validate allowed range. |
| grace_period_days | Number | Number of grace days. |
| status | Single select | Lease status. |
| template_ref | Single line text | Optional template reference. |

## 8. Table: LeaseSigner

Track required lease signers.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| signer_id | Autonumber | Record identifier. |

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| lease_id | Link to table → Lease | Required. |
| user_id | Link to table → User | Optional if signer has no registered account. |
| signer_name | Single line text | Required. |
| signer_email | Email | Required. |
| signing_status | Single select | Signing status. |
| signed_at | Date and time | Optional. |

## 9. Table: Document

Store document metadata and storage references.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| document_id | Autonumber | Record identifier. |
| owner_user_id | Link to table → User | Optional. |
| lease_id | Link to table → Lease | Optional. |
| document_type | Single select | Document category/type. |
| storage_ref | Single line text | File storage reference. |
| file_name | Single line text | Required. |
| mime_type | Single line text | Optional MIME type. |
| created_at | Date and time | Creation time. |

## 10. Table: Invoice

Invoices associated with leases.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| invoice_id | Autonumber | Record identifier. |
| lease_id | Link to table → Lease | Required. |
| period_start | Date | Billing period start. |
| period_end | Date | Billing period end. |
| issue_date | Date | Issue date. |
| due_date | Date | Due date. |
| subtotal | Number | Allow decimal values. |

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| fees | Number | Allow decimal values. |
| amount_due | Number | Allow decimal values. |
| amount_paid | Number | Allow decimal values. |
| status | Single select | Invoice status. |

## 11. Table: PaymentTransaction

Track payment attempts, failures, retries and partial payments.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| payment_id | Autonumber | Record identifier. |
| invoice_id | Link to table → Invoice | Required. |
| amount | Number | Allow decimal values. |
| currency | Single line text | Currency code; confirm project currency. |
| status | Single select | Payment status. |
| gateway_ref | Single line text | Optional gateway reference. |
| idempotency_key | Single line text | Required; enforce/validate uniqueness. |
| event_id | Single line text | Optional; deduplicate external events. |
| created_at | Date and time | Transaction creation time. |

## 12. Table: LedgerEntry

Financial ledger records.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| ledger_entry_id | Autonumber | Record identifier. |
| property_id | Link to table → Property | Required. |
| invoice_id | Link to table → Invoice | Optional. |
| payment_id | Link to table → PaymentTransaction | Optional. |
| entry_type | Single select | Ledger entry type. |
| amount | Number | Allow decimal values; agree sign convention. |
| posted_at | Date and time | Posting time. |
| reference | Single line text | Optional reference. |

## 13. Table: MaintenanceTicket

Tenant maintenance requests.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| ticket_id | Autonumber | Record identifier. |
| unit_id | Link to table → Unit | Required. |
| tenant_id | Link to table → User | Required. |

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| category | Single select | Maintenance category. |
| description | Long text | Required. |
| urgency | Single select | Urgency level. |
| status | Single select | Ticket status. |
| preferred_slots | Long text | Optional preferred appointment windows. |
| submitted_at | Date and time | Submission time. |

## 14. Table: TicketAttachment

Attachments linked to maintenance tickets.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| attachment_id | Autonumber | Record identifier. |
| ticket_id | Link to table → MaintenanceTicket | Required. |
| storage_ref | Single line text | File reference. |
| file_type | Single line text | File type. |
| file_size | Number | File size in bytes; validate positive value. |
| scan_status | Single select | Security scan status. |
| uploaded_at | Date and time | Upload time. |

## 15. Table: WorkOrder

Contractor work orders for maintenance tickets.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| work_order_id | Autonumber | Record identifier. |
| ticket_id | Link to table → MaintenanceTicket | Required. |
| contractor_id | Link to table → User | Required. |
| status | Single select | Work order status. |
| estimated_cost | Number | Optional; allow decimal values. |
| final_cost | Number | Optional; allow decimal values. |
| assigned_at | Date and time | Optional assignment time. |
| completed_at | Date and time | Optional completion time. |

## 16. Table: TicketStatusHistory

Append-only status history for maintenance tickets.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| history_id | Autonumber | Record identifier. |
| ticket_id | Link to table → MaintenanceTicket | Required. |
| changed_by | Link to table → User | Optional. |
| old_status | Single line text | Previous status; may be empty for first state. |
| new_status | Single select | New status. |
| note | Long text | Optional note. |
| changed_at | Date and time | Change time. |

## 17. Table: Notification

Notifications sent to users.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| notification_id | Autonumber | Record identifier. |
| recipient_user_id | Link to table → User | Required recipient. |
| category | Single select | Notification category. |
| channel | Single select | Delivery channel. |
| status | Single select | Delivery status. |
| related_record_type | Single line text | Type of related record. |
| related_record_id | Number | Related record ID; not a true foreign key. |
| created_at | Date and time | Creation time. |
| delivered_at | Date and time | Optional delivery time. |

## 18. Table: MessageThread

Conversation threads.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| thread_id | Autonumber | Record identifier. |
| property_id | Link to table → Property | Optional property context. |
| subject | Single line text | Optional subject. |
| created_by | Link to table → User | Thread creator. |

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| created_at | Date and time | Creation time. |

## 19. Table: Message

Messages inside conversation threads.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| message_id | Autonumber | Record identifier. |
| thread_id | Link to table → MessageThread | Required. |
| sender_user_id | Link to table → User | Required sender. |
| body | Long text | Required message body. |
| created_at | Date and time | Creation time. |
| edited_at | Date and time | Optional edit time. |

## 20. Table: AuditEvent

Audit trail for security-sensitive and business-critical events.

| Field name | Baserow field type | Implementation notes |
|---|---|---|
| audit_id | Autonumber | Record identifier. |
| actor_user_id | Link to table → User | Optional actor for system-generated events. |
| event_type | Single line text | Event type. |
| entity_type | Single line text | Affected entity type. |
| entity_id | Number | Affected entity ID. |
| occurred_at | Date and time | Event time. |
| correlation_id | Single line text | Optional correlation/trace ID. |
| metadata | Long text | Optional metadata; JSON may be stored as text. |

# 2. Relationship Register

Create each relation below as a Baserow Link to table field. Baserow may create a reciprocal field automatically; keep it if useful and avoid duplicate relationships.

| Source table | Link field | Target table |
|---|---|---|
| User | role_id | Role |
| Property | owner_id | User |
| Property | manager_id | User |
| Unit | property_id | Property |
| ListingMedia | unit_id | Unit |
| ListingMedia | uploaded_by | User |
| ApplicantEnquiry | unit_id | Unit |
| Lease | unit_id | Unit |
| Lease | tenant_id | User |
| LeaseSigner | lease_id | Lease |
| LeaseSigner | user_id | User |
| Document | owner_user_id | User |
| Document | lease_id | Lease |
| Invoice | lease_id | Lease |
| PaymentTransaction | invoice_id | Invoice |
| LedgerEntry | property_id | Property |
| LedgerEntry | invoice_id | Invoice |
| LedgerEntry | payment_id | PaymentTransaction |
| MaintenanceTicket | unit_id | Unit |
| MaintenanceTicket | tenant_id | User |
| TicketAttachment | ticket_id | MaintenanceTicket |
| WorkOrder | ticket_id | MaintenanceTicket |
| WorkOrder | contractor_id | User |
| TicketStatusHistory | ticket_id | MaintenanceTicket |
| TicketStatusHistory | changed_by | User |
| Notification | recipient_user_id | User |

| Source table | Link field | Target table |
|---|---|---|
| MessageThread | property_id | Property |
| MessageThread | created_by | User |
| Message | thread_id | MessageThread |
| Message | sender_user_id | User |
| AuditEvent | actor_user_id | User |

# 3. Validation and Open Decisions

### Document integrity

At least one of Document.owner_user_id or Document.lease_id must be populated. Field definitions alone do not enforce this.

### Role model

Current ERD assumes one primary role per user. If users may hold multiple roles, confirm and redesign before implementation.

### Lease model

Current ERD assumes one tenant per lease. Co-tenants/guarantors are an open scope decision.

### Property management

Current ERD supports one owner and one manager link per property. Confirm if multiple managers/delegated ownership are required.

### Notifications

related_record_type + related_record_id is a generic reference, not a true table relationship.

### Financial integrity

Confirm currency, amount sign conventions, partial payment behavior, uniqueness rules and ledger reconciliation.

### Status values

Agree allowed options for every Single select field before loading real data.

### Audit/history

TicketStatusHistory and AuditEvent should be append-only by process/permissions where feasible.

### Files

storage_key/storage_ref fields store references; secure storage and access rules are configured separately.

### Testing

Create sample records to test Role → User → Property → Unit → Lease → Invoice → PaymentTransaction and Unit → MaintenanceTicket → WorkOrder.

# Recommended build order

Role → User → Property → Unit → ListingMedia / ApplicantEnquiry → Lease → LeaseSigner / Document → Invoice → PaymentTransaction → LedgerEntry → MaintenanceTicket → TicketAttachment / WorkOrder / TicketStatusHistory → Notification / MessageThread → Message → AuditEvent.

This is a practical build order based on table dependencies. It does not introduce new entities or change the source ERD.